

Lab-based practical – 5COSC019W Object Oriented Programming

The following Java-based system for managing a library centre is provided. You will have about ten days to study and analyse the project. You are expected to familiarise yourself with the code and the implemented classes. The system has been provided as a NetBeans project, and it includes the following classes:

- **LibraryCentre_MockLabBasedPractical.java** (main() method is here)
- **Item.java**
- **Book.java**
- **DVD.java**
- **WestminsterLibraryManager.java**
- **LibraryManager.java**
- **ItemTableGUI.java**
- **ItemTableModel.java**

The project files are attached in a zip folder.

The system is represented in the **UML Class diagram provided below** in this specification.

The UML Class diagram can be used by the student as a reference and a map of the system.

In this lab-based practical assessment, your task is to extend the functionality of this Java project by implementing additional Java code.

Instructions Before Starting

- 1) Unzip the folder and open the project in NetBeans.
- 2) Once opened, locate the **LibraryCentre_MockLabBasedPractical** class and, in the comments section at the beginning of the file, add your first name, last name, and student ID.

Instructions for the Submission

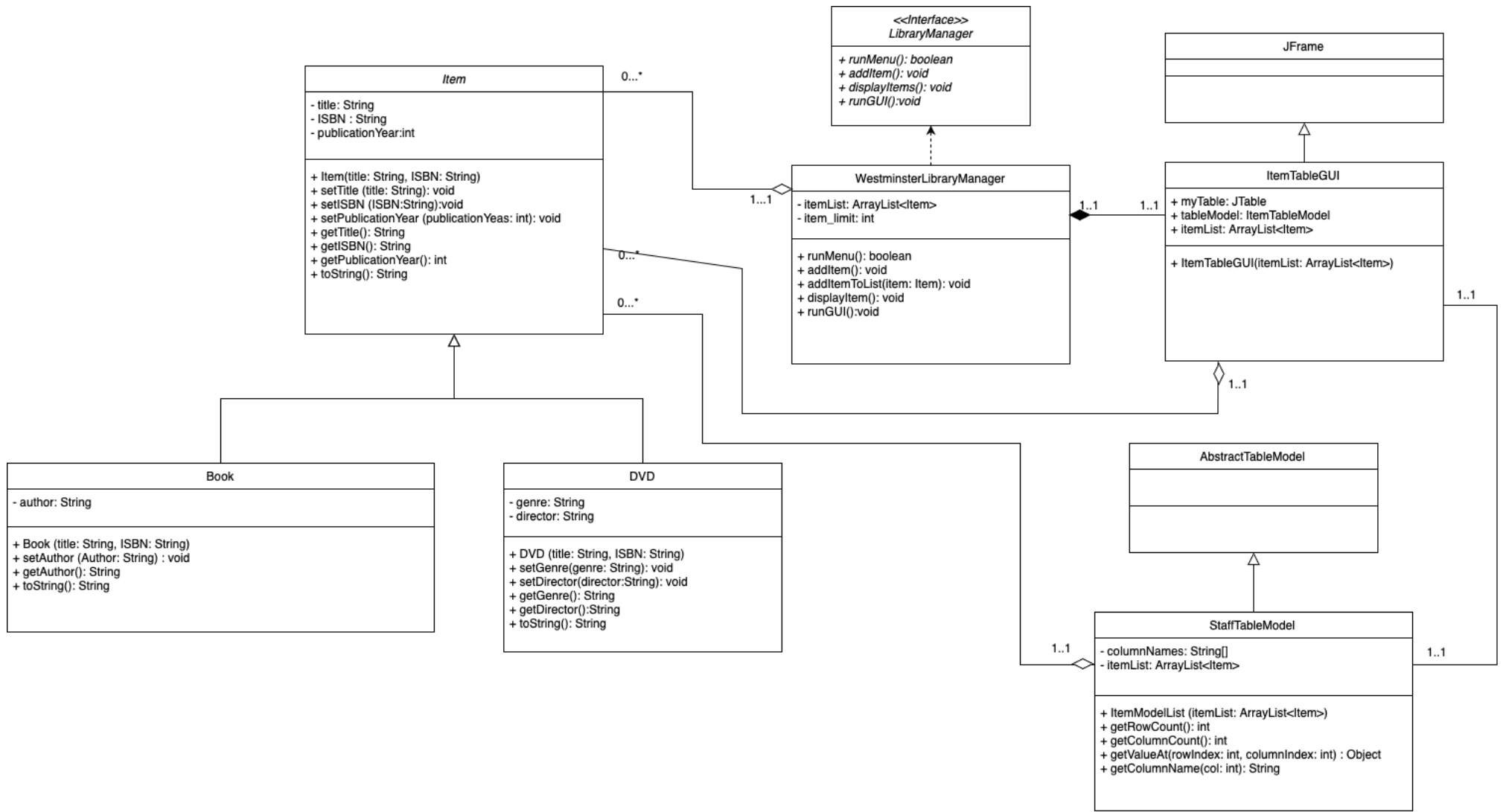
- 1) Save your NetBeans Project files. THIS is VERY IMPORTANT, OTHERWISE YOU MIGHT ZIP A BLANK PROJECT!!!
- 2) Zip the whole NetBeans project and submit the zip file.
- 3) Select which task has been attempted or not on BB.
- 4) Once you have submitted, download the project and check that it contains all the edited files.

If you do not know how to zip the NetBeans project, you will find a guide at the end of this document.

(NOTE: During the real lab-based practical test, **2 marks will be deducted** if you require assistance from the tutor to create (zip) your file.)

You are allowed to access the module materials (lectures and tutorials) on BB, but you are not permitted to bring your own notes or books. In addition, only one NetBeans project — the one related to the test — may be open in your IDE.

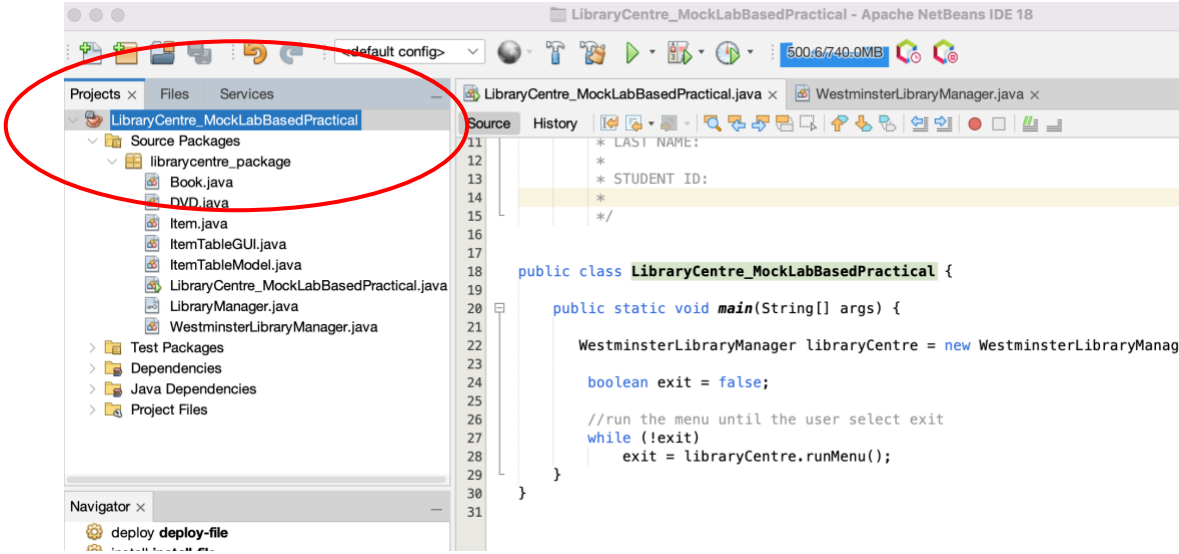
Students are not allowed to use any other websites. If you use any Generative AI tools, such as ChatGPT, the mark will be 0, and you will be reported for misconduct.



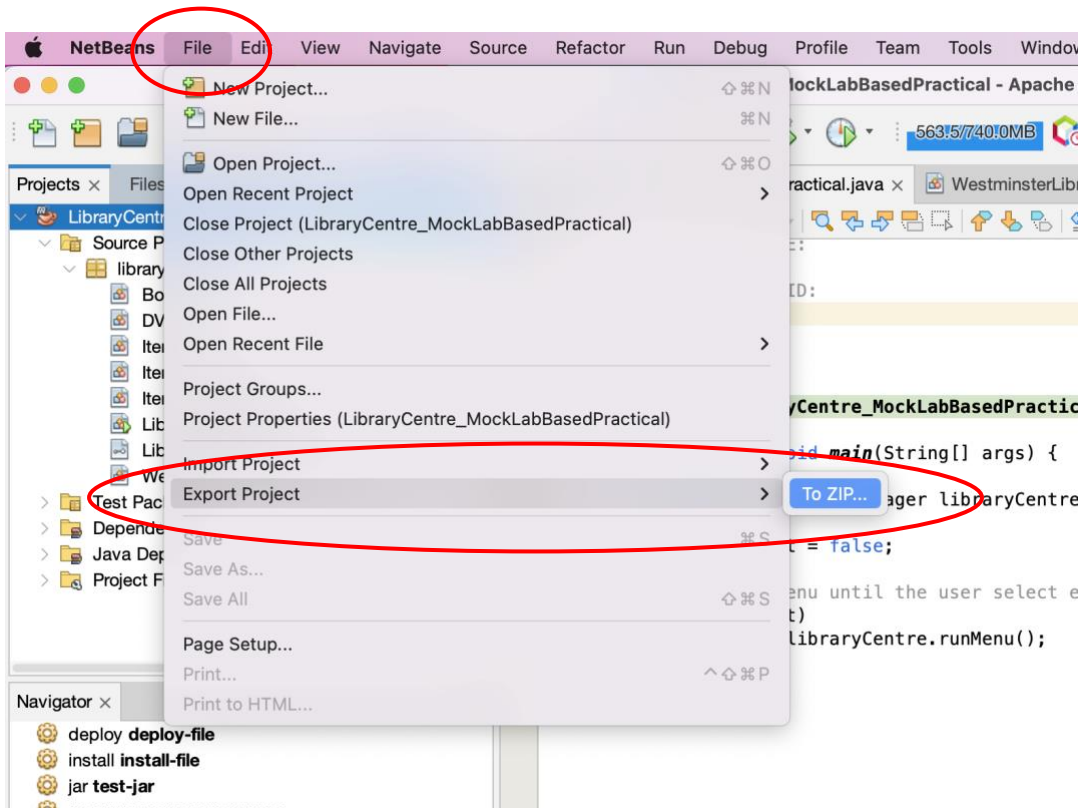
How to Zip NetBeans project

On the left side of your NetBeans window, go to the **Projects** tab.

Select the project that you want to zip. It will be highlighted in blue as shown in the following image:



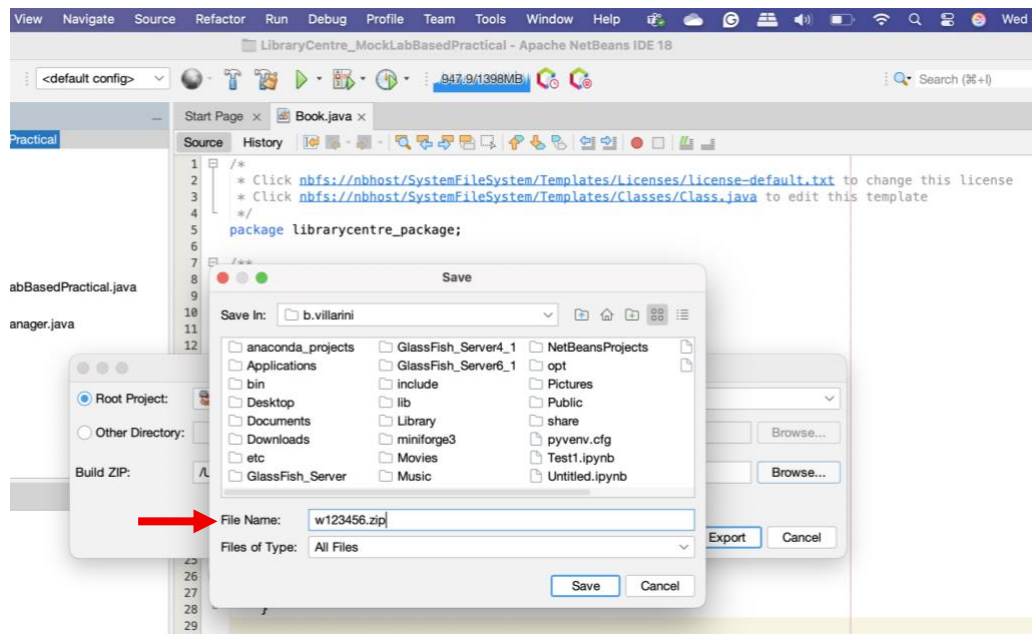
Keeping the project selected as shown, go to the above tab **File** of NetBeans IDE, and then go to **Export Project** and select **To ZIP...** , as shown below:



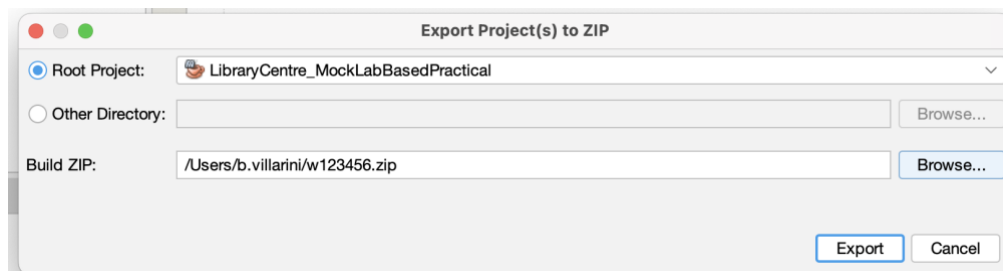
Once you click on **ToZip...** , the following window will open:



You need to change the path where you want to save your zip file. Choose a folder you are familiar with and know how to access. Then, add a file name (for example, your student ID) followed by **.zip**.



Click “Save”. You will see this window:



Now you can click the **Export** button.

Your zip file is inside the folder you selected.

MOCK TEST - Lab-based practical – 5COSC019W Object Oriented Programming

TASK 01 (25 marks)

1. Create a new Java class called "Magazine" within the Package librarycentre_package (2 marks)
2. Ensure the class inherits from "Item" class and follows the principle of inheritance. (4 marks)
3. The **Magazine** class should have the following instance variables:
 - o **Title, publication year** and **ISBN**, which are inherited from the **Item** class.
 - o Additionally, include **Issue Number, publication frequency** (e.g., weekly, monthly, quarterly) and the **editor**. (4 marks)
4. Implement a **constructor** in the Magazine class that takes the **title** and **ISBN** as parameters. (4 marks)
5. Ensure the class includes appropriate **setter and getter methods** for all the instance variables to follow the encapsulation principle (validation is not necessary). (8 marks)
6. **Override the toString method** to return a String representing a description of the magazine object. (3 marks)

TASK 02 (24 marks)

1. In **LibraryManager** Interface add an abstract method called **editTitleItem (2 marks)**. This method should have the following signature:
`void editTitleItem()`
2. In the **WestminsterLibraryManager** class, you should override the method **editTitleItem()** that you just added in the Interface. (2 marks).
3. The method editTitleItem() in the **WestminsterLibraryManager** should ask the user to insert from the keyboard the ISBN of the item to be edited (3 marks).
4. If it is found, this method should print the following:
 - The current **Title** , and **publication year** (4 marks) of the searched item;
 - the type of the item, whether it was a **book or a DVD** (3 marks).
5. The method should ask the user to insert the new **Title** (2 marks).
6. The new title for the selected item should be saved correctly. (2 marks)
7. Modify the **runMenu()** method in the **WestminsterLibraryManager** class to include this functionality in the console menu (6 marks).

Task 03 (17 marks)

1. In the **WestminsterLibraryManager** class, within the **displayItems** method, add the necessary code to display the list of items sorted by publication year (4 marks).
2. Utilise the appropriate methods from the **Collections** class to perform the sorting (5 marks).
3. Modify the **Item** class to implement the **Comparable** interface so that it supports sorting by publication year (8 marks).

Task 04 (10 marks)

1. Modify the **ItemTableModel.java** class to add an additional column displayed on the table that shows the **Type** of the Item: if it is a book, or a dvd(, or a magazine (if implemented the class magazine)10 marks).

Task 05 (15 marks)

1. Add an **event handler** to the "Info" button in the **ItemTableGUI.java** class (8 marks).
When clicking the button, the user should see the **total number of books, the total number of DVD and the total number of magazines (7 marks) displayed on the GUI.**

Task 06 (9 marks)

1. Create a **unit test** class using the **JUnit framework** integrated into NetBeans (1 marks).
2. The unit test should verify the functionality of **AddItemsToList()** method implemented in the **WestminsterLibraryManager** class (8 marks):